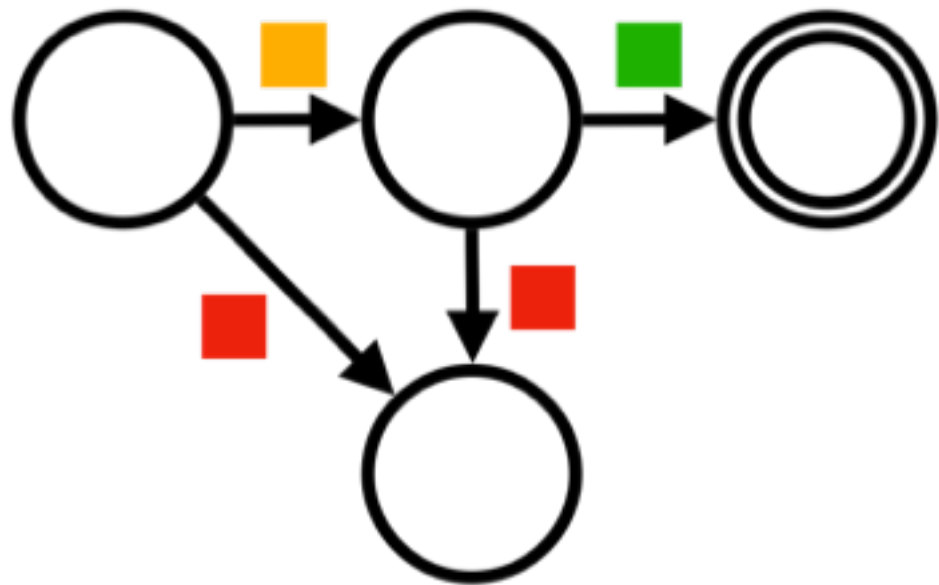




Automatic as an instructional modality

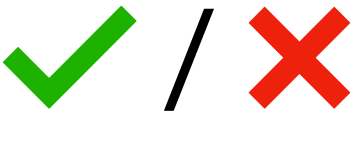
• **Unambiguous, unlikel natural language**

• Has well-defined semantics

• **Compositional objects**

Pick up the laundry
and fold it. Avoid
breaking anything!







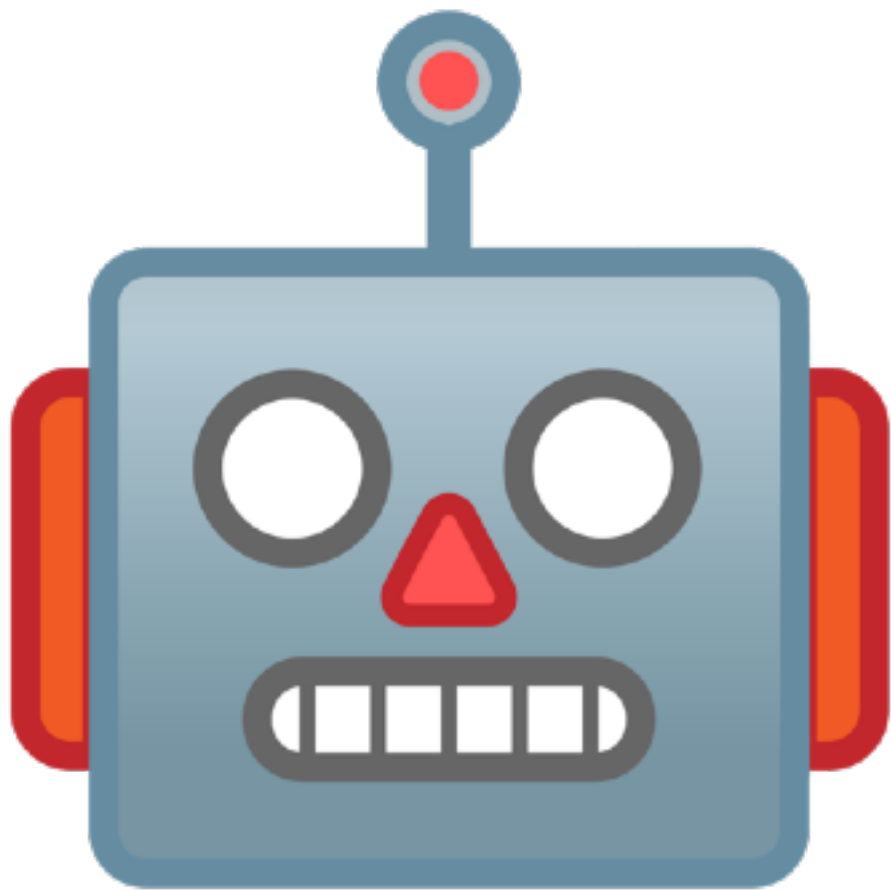
Jothimurugan, Kishor, Rajeev Alur, and Osbert Bastani. "A composable specification language for reinforcement learning tasks." Neurips 2019.

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Shah, Ameesh, et al. "LTL-Constrained Policy Optimization with Cycle Experience Replay." TMLR 2025.





Our goal is to compare the two modalities.



and for it. Avoid

Pick up the laundry

breaking anything!







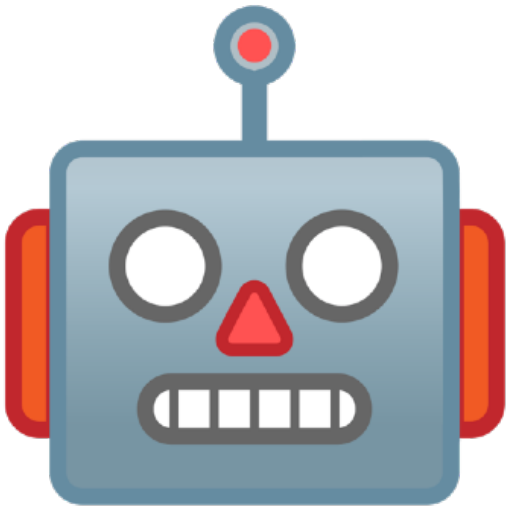
Hassanbeig, Mohammadsein, Daniel Koenig, and Alessio Abate. 'Deep reinforcement learning with temporal logic.' *ECRMA* 2020.

Learte, Riggo Toro, et al. 'Reward machines: Exploiting reward function structure in reinforcement learning.' JAI R 2022.

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Yalcinkaya, Beyazit, et al. 'Compositional autodecoding for al-condition reinforcement learning' Neurips 2024.





RAD *Embeddings*



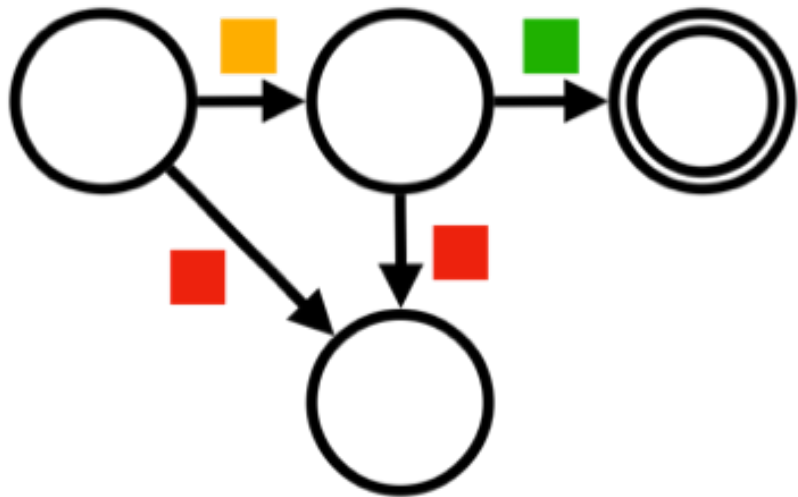
Qwen3-Embedding





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Zhang, Yanzhao, et al. 'Qwen3: Embedding: Advancing text embedding and reranking through foundation models.' arXiv 2025.



and for it. Avoid

Pick up the laundry

breaking anything!

